

Frequency Drone

Manual for the Deluge with firmware 1.2.1-mastertune v16 (v15: without drone tracks; v13 and v14: also without Life, FM and Pulse; v12: also without triplets, Attack/Release, knob sections and reverb send, and its pads pulse)

The drone plays up to **16 sustained tones** under the music of your song. Each tone can be steady or carry a beat: **binaural**, **monaural** or **isochronic**, free in hertz or synced to the tempo, also in triplets. You set the pitch in hertz or as a note that follows the Master Tune. Isochronic pulses have attack and release. The drone ducks under the song's sidechain, can send to the song reverb and is saved with the song.

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Start quiet. Sustained tones, especially low ones and many at once, can seem louder than expected on headphones. Start at a low volume and raise it slowly.

Not a therapy: The names Delta to Gamma only describe frequency ranges. How beats affect people is scientifically disputed. The drone is a sound tool, not a medical device.

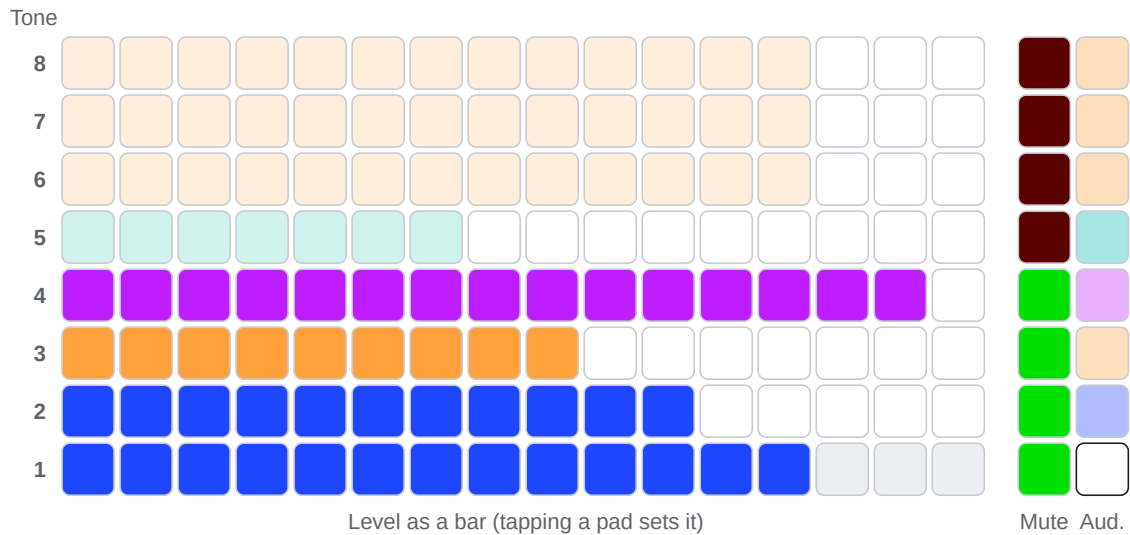
1. Quick start

1. In **song view**, press the **SCALE** button. The drone view opens and the Scale LED lights up.
2. In the bottom row, press the **mute pad** (first column to the right of the grid). It turns green and tone 1 sounds: a sine at 200 Hz.
3. Turn the **upper gold knob** to change the pitch: 1 Hz per detent, 10 Hz when turned fast.
4. Press the **KIT** button: the tone becomes *binaural*. On headphones you now hear a beating of 10 Hz.
5. Turn the **lower gold knob** to change the beat, or **press** it so the beat follows the tempo (1/16 at first).
6. Go back to song view with **BACK**, **SONG** or **SCALE**. The drone keeps sounding.

Alternatively, open the drone view in song view by pressing **SELECT** (song menu) → *Drone*. The arranger does not have this menu item, because the way back always leads to song view.

2. The drone view

The view is built like song view: one row per tone instead of one per clip, **tone 1 at the bottom**. Eight tones are visible at once. Use the **Y encoder** (vertical) to scroll to tones 9–16.



The pads of the drone view. Each row is a tone. The bar shows the level in the mode's color, followed on the right by the mute column and the audition column.

Pads in a row

- **Bar** = level of the tone, 16 pads for 0–50.
- **Tapping a pad** sets the level and selects the tone: pad 1 = 3, pad 8 = 25, pad 16 = 50.
- **Selected tone:** its row glows faint gray behind the bar.
- **Tones that are off** are strongly dimmed; their settings are kept.
- **Steady display:** the pads light evenly and do not flash with the beat.

Colors and side columns

- **Orange:** Tone (steady)
- **Blue:** Binaural
- **Turquoise:** Monaural
- **Violet:** Isochronic
- **Mute column:** ■ green = on, ■ dark red = off. Tapping toggles the tone and selects it.
- **Audition column:** tapping selects the tone. White = selected, otherwise the mode color, dimmed.

3. All controls

Control	Effect
Pad in the grid	set the level of this row's tone and select the tone
Mute pad	tone on/off, select tone
Audition pad	select tone
Turn Y encoder	scroll: tones 1–8 to 9–16
Turn X encoder	level of the selected tone in steps of 1 (0–50)
SHIFT + X encoder	pan of the selected tone (–32 left to +32 right)
Turn Select	pitch of the selected tone in tactile detents: in Hz 1 Hz per click, with SHIFT 0.1 Hz; as a note one semitone, with SHIFT one cent
SYNTH / KIT / MIDI / CV	mode of the selected tone: Tone / Binaural / Monaural / Isochronic. The button's LED shows the current mode.
Press Select	open the menu of the selected tone and of the whole drone (chapter 11)
Gold knobs in the Cutoff/Resonance section (pitch/beat, selected on opening)	
Turn upper gold knob, set in Hz	frequency: 1 Hz per detent; 10 Hz when turned fast; with SHIFT 0.01 Hz
Turn upper gold knob, set as a note	note in semitones; with SHIFT cents (±50)
Press upper gold knob	switch between Hz and note. The pitch stays (the nearest note plus cents, or the note's frequency).
Turn lower gold knob, beat free	beat: 0.1 Hz per detent, with SHIFT 0.01 Hz (0–100 Hz)
Turn lower gold knob, beat synced	note value: 1/1, 1/1T, 1/2, 1/2T ... 1/64, 1/64T (T = triplet)
Press lower gold knob	tempo sync on/off. Switching it on brings back the last note value used (1/16 at first). Switching it off brings back the beat in Hz.
Gold knobs in the other sections (press the mod button next to the knobs)	
Level/Pan	upper: level of the selected tone (0–50) · lower: pan (–32 to +32)
Attack/Release	upper: attack · lower: release of the selected tone's isochronic pulse (0–50, shown as 0–100 %)
Sidechain/Reverb	upper: sidechain ducking of the whole drone (0–50) · lower: reverb send of the whole drone (0–50)
Delay (v15)	upper: Life of the whole drone (0–50) · lower: Rate, how fast everything moves (0–50, 25 = default)
ModFX (v15)	upper: FM of the whole drone (0–50), press : FM form Sine or Saw · lower: Pulse width (5–50 %)
other mod buttons	no function; the section stays
BACK , SONG , SCALE	back to song view
PLAY , RECORD , tempo, save, load	as usual

The LED of the selected mod button lights up, and the knobs' LED rings show the values (for pitch/beat they stay dark). Pressing the gold knobs only works in the Cutoff/Resonance section and (upper knob) in the ModFX section. The selected section stays,

even after the menu. In the drone view, the gold knobs act only on the drone, not on the song parameters. There is no undo in the drone view; drone settings are not part of the undo history.

4. Display: OLED and 7-segment

OLED

- **Title:** number and mode of the selected tone, e.g. 3: Binaural. If the tone is off, - off follows.
- **Large:** the pitch, 200.00 Hz or as a note, e.g. A2 +12c.
- **Bottom line:**
 - Tone mode: timbre and level, e.g. Sine Level 40
 - other modes: beat with frequency band, e.g. Beat 10.00 Hz Alpha
 - with tempo sync: 1/16: 8.00 Hz Alpha, as a triplet 1/16T: 12.00 Hz Beta
- Short pop-ups show level, pan, mode, Hz/Note, SYNC/Beat, the selected knob section and its values, e.g. Attack 12%.

Note names follow the Deluge numbering: A3 = 440 Hz, middle C = C3.

7-segment

- **Frequency** with as many decimal places as fit: 55.25, 440.0, 1200.
- **Note** as a name, e.g. A2 (sharp shown as a dot).
- **While turning**, shown briefly:
 - Level: number
 - Pan: number
 - Beat: in tenths, e.g. 7.8
 - Note value: 16 for 1/16, 16T for the triplet

5. The four modes

Binaural

f = 200 Hz, Beat = 10 Hz

left 195 Hz

right 205 Hz

beating arises in the head

Monaural

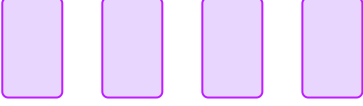
both tones in both ears



beating audible in the sound

Isochronic

tone pulses with the beat



on/off with attack and release

Mode	What you hear	Remark
Tone	A steady sustained tone, the same left and right (mind the pan).	Beat and sync play no role.
Binaural	Left the frequency minus half the beat, right plus half the beat. The beating only arises in the head.	Works only with headphones ; over speakers it sounds like Monaural.
Monaural	The same two tones, but both in both ears. The beating is audible in the sound itself, also over speakers.	Good for rooms and recordings.
Isochronic	One tone that switches on and off with the beat. Attack and release set how softly it starts and stops.	Clearest effect, also over speakers.

Attack and release of the isochronic pulse

The pulse is on for half of each beat cycle. **Attack** is the time in which it rises at the start, **Release** the time in which it then fades. Both go from 0 to 50. Each step is 2 % of half the cycle length, so 50 is the whole half that is on. This

way the pulse keeps its shape when the beat or tempo changes. An edge is never shorter than 1 ms, so nothing clicks. Both start at 6: at 10 Hz that is 6 ms, as in v12.

Attack / Release	Sound
0–6	clear, hard pulse
15–25	soft, without edges
50 / 50	almost sinusoidal swelling and fading
50 / 0	swells and cuts off, like a tone played backwards

A new mode or a new timbre is faded briefly through silence, so nothing clicks. Level, pitch, beat and pan glide smoothly to the new value.

Band	Delta	Theta	Alpha	Beta	Gamma
Beat	below 4 Hz	4 to below 8 Hz	8 to below 13 Hz	13 to below 30 Hz	30 Hz and above

The OLED shows the band for the current beat. The beat can be at most as high as the frequency itself, so the lower tone does not fall below 0 Hz.

6. Pitch: hertz or note

In hertz

10.00 to 5000.00 Hz, accurate to 0.01 Hz. Useful for fixed frequencies, for example 432 Hz, 528 Hz or a room resonance. A frequency in hertz stays fixed, even if you change the Master Tune.

If you switch between Hz and note with a knob press or in the menu (*Pitch as*), the pitch is kept. A frequency becomes the nearest note with cents; a note becomes its frequency at the current Master Tune.

As a note

Note 0–127 plus –50 to +50 cents. The tone **follows the Master Tune** (Settings → Tuning) like all other sounds of the Deluge. If you tune the Master Tune to an external instrument, the drone matches automatically.

7. Beat and tempo sync

The beat is the beating or pulse frequency of the Binaural, Monaural and Isochronic modes. Free, it runs from 0 to 100 Hz in 0.01 Hz steps; it starts at 10 Hz. With tempo sync it follows the song tempo, straight or as a triplet (three in the time of two, so 1.5 times as fast):

Note value	1/1	1/2	1/4	1/8	1/16	1/32	1/64
Beat at 120 BPM	0.5 Hz	1 Hz	2 Hz	4 Hz	8 Hz	16 Hz	32 Hz
Band	Delta	Delta	Delta	Theta	Alpha	Beta	Gamma
Triplet at 120 BPM	0.75 Hz	1.5 Hz	3 Hz	6 Hz	12 Hz	24 Hz	48 Hz

- **When the song is playing**, the pulse locks onto the bar grid (accurate to 1.2 ms in the test). This also applies with external MIDI clock and with sync scaling.
- **When the song is stopped**, the beat runs on freely at the set tempo.
- The beat follows **tempo changes** smoothly.

8. Timbres

Timbre	Sound
Sine	pure sine, clearest for beats
Soft	soft harmonics 2–5, warm
Organ	octaves like organ drawbars
Rich	harmonics 2–8, sawtooth-like and bright
Pulse (v15)	square wave with adjustable width (<i>Pulse width</i> 5–50 %, default 30): hollow and nasal when narrower, full at 50 %. With Life the width drifts slowly.

All timbres are band-limited: harmonics above 18 kHz are dropped, so nothing aliases or chirps on high tones. You choose the timbre in the menu (*Timbre*). Pulse applies to the single tone, the width to the whole drone.

8a. Life and FM (v15)

A sustained tone soon sounds rigid. With **Life**, each tone moves by itself, as if played by hand; with **FM** it gets a color that blooms and fades again. Both apply to the whole drone, but each tone moves on its own paths.

Setting	Effect
Life 0–50	Each tone drifts on slow random paths that never repeat. At 50: pitch ± 6 cents (path at about 0.08 Hz), volume ± 3 dB ("breath"), brightness and stereo position ("shimmer"). Everything grows evenly with Life. On a beating tone, both sides drift together: the beat stays exactly as set.
Rate 0–50	How fast Life and the blooming of the FM move. 25 as designed; every 12.5 steps doubles or halves the speed.
FM 0–50	A modulator at the tone's frequency plus 0.3 Hz colors it. The depth blooms and fades, each tone on its own path. On high tones it is limited, so nothing audibly aliases.
FM form	Sine: soft, bell-like · Saw: brighter, more tinny. Switching crossfades.

- **At Life 0 and FM 0** (and without Pulse) the drone sounds exactly as in v13 and v14.
- Changes to Life, Rate and FM glide; nothing clicks. A switch to or from Pulse goes through silence, like any change of timbre.
- Life only moves what the drone itself does. Tempo sync, sidechain and Master Tune still apply as set.

9. Level, pan, volume and mix

- **Level** per tone, 0–50 in 1 dB steps, 0 is silent. A tone at level 50 with drone volume 50 sits at -12 dBFS. New tones start at level 40.
- **Pan** per tone, -32 to $+32$, equal power: 0 dB in the center, up to $+3$ dB on one side at the far ends. With Binaural, pan acts as a balance between the ears: panned hard, one ear barely hears its tone and the beating is lost.
- **Volume** (in the menu): volume of the whole drone, 0–50 in 1 dB steps, 50 = 0 dB, starts at 40.
- **Song volume**: the drone follows it, like the metronome.
- **In the mix** the drone is added after the song effects, so without delay and song filter. It can be heard over USB audio (v8).
- **Reverb** (menu *Reverb amount* or lower gold knob in the Sidechain/Reverb section): send of the whole drone into the song's reverb, 0–50. At 50 it sends as much as a sound with the reverb knob turned fully up. The send follows level, pan and sidechain ducking and glides smoothly when turned.

Many loud tones together can clip: 16 tones at level 50 add up to far more than full scale. For pad sounds of many tones, lower the level per tone or the drone volume.

10. Sidechain

The drone can duck under the song's sidechain triggers, for example under a kick with *Send to sidechain*. The level glides from sample to sample, so nothing clicks.

Menu item (Sidechain)	Effect
Volume ducking	Amount 0–50, off at 0. At the lowest point of a full hit, the level drops linearly: 10 → -2 dB, 20 → -4 dB, 30 → -8 dB, 40 → -14 dB, 50 → fully gone. While turning, the OLED shows the depth, e.g. Sidechain 30 -8 dB.
SYNC	envelope in tempo, as with the other sidechains
ATTACK / RELEASE	how fast the drone ducks and comes back
SHAPE (title <i>Sidech. shape</i>)	shape of the curve, as with the reverb; only visible when Volume ducking is above 0

11. The menu

Pressing **SELECT** opens the *Drone tone* menu for the selected tone. Items that do not apply at the moment are hidden. The names are written as the Deluge shows them.

Item	Values	Visible
MODE	Tone, Binaural, Monaural, Isochronic	always
Pitch as	Hz, Note (switches at the same pitch)	always
Frequency	10.00–5000.00 Hz	with Hz
Note	0–127, shown as a note name	with Note
Cents	–50 to +50	with Note
Beat	0.00–100.00 Hz	not with Tone, and sync off
SYNC	Off, 1/1, 1/1T to 1/64, 1/64T	not with Tone
ATTACK	0–50 (shown as 0–100 %)	with Isochronic
RELEASE	0–50 (shown as 0–100 %)	with Isochronic
Timbre	Sine, Soft, Organ, Rich, Pulse	always
Level	0–50	always
PAN	–32 to +32	always
Volume	0–50, whole drone	always
Reverb amount	0–50, send of the whole drone	always
Life	0–50, whole drone (chapter 8a)	always
Rate	0–50, whole drone, 25 = default	always
FM	0–50, whole drone	always
FM form	Sine, Saw	always
Pulse width	5–50 %, whole drone	always
Sidechain	submenu (chapter 10), whole drone	always

BACK leaves the menu and takes you back to the drone view.

12. Saving, loading, export

- **Saving:** the drone is saved with the song (tag <drone>), but only if it has been set up. Older firmware simply skips it.
- **Loading:** when you load another song or use *Clear Song*, the drone fades out in about 60 ms. The new song's drone starts from silence.
- **Stem export:** the drone is not in the stems. Afterwards it starts again from silence.
- **Recording:** you record it with the rest of the song via USB audio or the output.

13. With external devices

Set the tones to **Note** and tune the **Master Tune** (Settings → Tuning) to your device, for example a volca keys. Then the drone and the instrument share the same tuning. If you move the Master Tune later, all tones set as notes follow; tones in Hz stay fixed.

14. Recipes

Goal	How to
Calm alpha bed	Tone 1: Binaural, Sine, 200 Hz, Beat 10 Hz, Level 35. Tone 2: Binaural, Soft, 300 Hz, Beat 10 Hz, Level 28. Listen on headphones.
Pulse in the groove	Isochronic, Sync 1/16 (8 Hz at 120 BPM), Rich, one octave below the song's root note. Add sidechain with Volume ducking 20.
Soft triplet pulse	Isochronic, Sync 1/8T, Soft, Attack 20, Release 30. Shape it live with the gold knobs in the Attack/Release section.
Drone in the room	Monaural, Organ, Reverb amount 20–30: the drone sits in the same room as the song.
Root-note pad	Three tones as notes: root, fifth, octave; Timbre Organ, mode Tone, Level 30–35. The tones follow the Master Tune.
Floating choir	Monaural, Soft, the same note on two tones, Beat 0.3 and 0.5 Hz, Pan –20 and +20.
Slow movement	Binaural or Monaural with Sync 1/1 or 1/2: one beating per bar or half bar.
Breathing pad (v15)	Three Soft tones as root, fifth, octave; Life 25, Rate 20. The pad moves slowly by itself, each tone a little differently.
Bells in the fog (v15)	Monaural, Sine, two high tones; FM 30 with form Sine, Life 15, Reverb amount 25.
Rough bass drone (v15)	Tone, Pulse, one low note; Pulse width 15, FM 20 with form Saw, Life 10. Put it under the kick with sidechain.

15. Specifications and limits

Property	Value
Tones	16, each with mode, timbre, pitch, beat, sync, level, pan
Frequency / note	10.00–5000.00 Hz · note 0–127 ± 50 cents
Beat	0–100 Hz free (accurate to 0.01 Hz) or 1/1–1/64 synced, also as triplets
Isochronic pulse	Attack and release 0–50 (each step 2 % of half the cycle length), at least 1 ms
Reverb send	0–50 for the whole drone
Level, Volume, Pan	0–50 (1 dB), 0–50 (1 dB), –32 to +32
Purity	Sine: spurious products below –108 dB
Life, FM, Pulse (v15)	Life and FM 0–50 and Rate 0–50 for the whole drone, FM form Sine or Saw, pulse width 5–50 %
CPU time (Cortex-A9, counted in the emulator)	1 sine tone 0.35 %, 1 binaural tone 0.5 %, 16 binaural tones with harmonics 6.9 % of the CPU. With v15 per binaural tone: Life 0.7–1.0 %, FM 1.3 %, Pulse 1.0 %, all together 2.1 %; 16 tones with everything just over 30 %. The drone is never culled like synth voices.
Memory	Wavetables 27 KB internal RAM, with v15 another 14 KB (sawtooth tables for Pulse and FM)

- **Tested** in the emulator (46 checks on PC and Cortex-A9) and on the device with v12. The new features of v13 and v15 are not yet tested on the device; Life, FM and Pulse are checked on the PC with 116 checks.
- The song drone sends no MIDI and takes no notes from the keyboard or MIDI. Drone tracks (chapter 16) take notes: they open the row's gate.
- The song drone's values cannot be automated. With drone tracks, the pitch of each row is recorded with REC as a Hz track (chapter 16).

16. Drone tracks (v16)

The song drone sounds as long as it is on. With **drone tracks** you place drones where you want them: a drone track is a kit whose rows are drone tones. You launch their clips in song view and place them in the arranger like any kit clip. Several drone tracks can overlap, each with its kit's effects, volume and sidechain.

What	How to
Make a drone track from the drone	In the drone view, SHIFT + KIT . This creates a new kit (DRONE1, DRONE2 ...) with 16 rows, the tones of the song drone; the tones that are on get a note across the whole clip. The clip sits at the bottom of song view and is not launched yet. The song drone itself stays unchanged.
Turn a kit row into a drone row	In a kit's clip view, hold the row's audition pad (or create a new row), then SHIFT + KIT . The row sounds at 200 Hz.
Gate	A drone row's notes are its gate: as long as a note lasts, the tone sounds, with a soft fade in and out. A note across the whole clip holds over the loop boundary. With the row's arpeggiator, the tone pulses in rhythm.
Select a row	Its audition pad (Affect Entire off). If the row is sounding from the sequencer at the time, the pad only selects it; it keeps sounding.
Pitch	Turn Select: 1 Hz per detent, with SHIFT 0.1 Hz. Upper gold knob: 1 Hz, 10 Hz when turned fast, with SHIFT 0.01 Hz. A tone set as a note: semitones, with SHIFT cents.
Level	Lower gold knob
Row menu	Press Select: mode, pitch, beat, sync, pulse, timbre, level, pan, plus the row's arpeggiator and randomizer
Display	OLED: mode and Hz as the row's name; 7-segment: the Hz

Recording and playing Hz changes

- **PLAY**, then **RECORD**, then turn Select or the upper gold knob: each change is written as a node into the row's **Hz track** and plays back in the next loop. Between two nodes the tone glides in about 15 ms. This also works when recording into the arranger.
- Without **RECORD**, the row's base pitch changes; a recorded Hz track shifts with it.
- While **RECORD** is on: **SHIFT** + Select records the fine 0.1 Hz steps. **SHIFT** + upper gold knob deletes the Hz track, like any automation; the row goes back to its base pitch at once.
- The Hz track belongs to the row, not to the notes: recording notes, moving them or spreading them in a Euclidean pattern leaves it unchanged.
- MIDI pitch bend on a drone row acts within the row's bend range (default 2 semitones) and is also recorded while **RECORD** is on.

Good to know

- CPU time: about 0.3–0.45 % CPU per sounding row, 16 rows 5–7 %. Rows with a closed gate cost nothing.
- A new drone kit sounds as loud as the song drone at the same setting.
- Older firmware opens songs with drone tracks without the drone rows; the other rows stay correct. If you save them there, the drone rows are gone.